TERRA TREND

NASA SPACE APPS CHALLENGE 2026 / EARTH SYSTEM TREND DETECTIVE

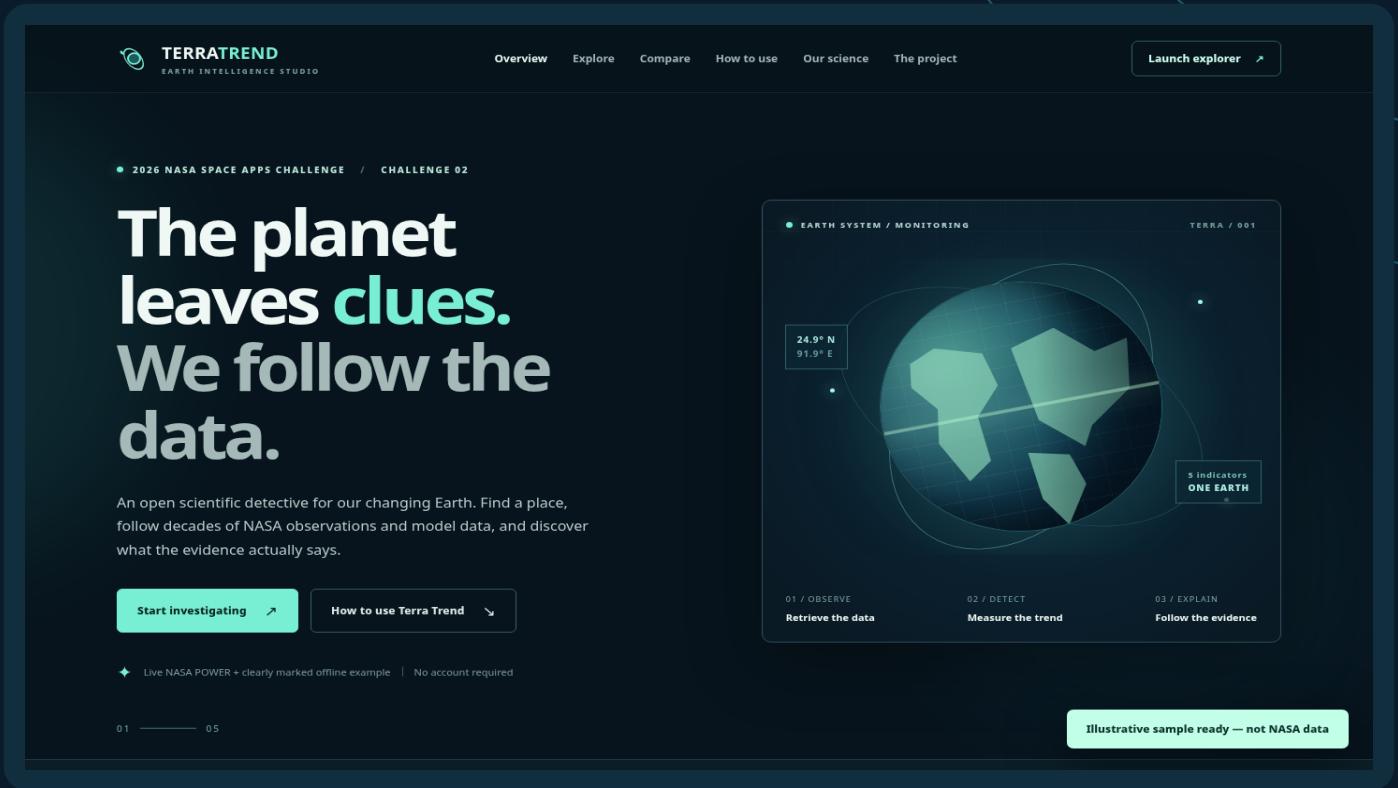
THE COMPLETE BEGINNER'S GUIDE

An A-to-Z handbook for the Terra Trend website

Every button. Every feature. Every graph.

Every scientific result, explained in simple language.

Plus your publishing and hackathon presentation guide.



VERSION | 07 NOVEMBER 2026

Based on the verified Terra Trend web files and test notes

START HERE / BEFORE ANYTHING ELSE

What is Terra Trend?

Terra Trend is a website that helps ordinary people investigate whether environmental measurements have changed over time in a chosen place. You choose a location, what to measure, and which years to study. The app prepares annual data, draws a graph, estimates a long-term trend, reports statistical evidence, and can compare two locations.

> **In one sentence:** Terra Trend turns a question such as "Has Sylhet been getting warmer since 1990?" into a repeatable data investigation that a non-scientist can explore.

Two modes - never confuse them

Mode	What happens	May it be used as NASA evidence?
Offline interactive example	The website generates **synthetic practice numbers**. It can work without internet.	**No.** These numbers are invented teaching examples.
Live NASA POWER	The website asks NASA POWER for monthly environmental estimates and analyzes them.	**Only after** the real data connection and source are verified.

What you will learn from this guide

- What each tab, button, dropdown, map, chart, and export feature does.
- How to choose the five environmental indicators and a study period.
- How to interpret slope, total change, p-value, confidence interval, and missing years.
- How the website uses NASA POWER data; what the data can and cannot establish.
- How to open, publish, check, explain, and submit the site for the challenge.

> **Current status (7 November 2026):** The site passed 44 automated checks, including controls, graphs, mathematics and simulated NASA responses. **A genuine live NASA request and the deployed public website have not yet been verified.** The guide does not present this as fully tested NASA research.

NAVIGATE THIS HANDBOOK

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The page numbers below update automatically. You can also use the PDF viewer search function for any term.

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Fast route: Read Sections 03, 04, 08, 09, 10 and 18 first. Those sections teach you how to operate the website and understand its main results.

01 / The challenge and our project

Why Terra Trend exists and how it relates to the NASA Space Apps challenge.

What NASA asks us to build

The 2026 challenge is titled **"Be An Earth System Trend Detective!"**. It asks for a tool that can select Earth-system variables measured by NASA missions or generated by Earth science models, display how those variables change over time, and investigate **what changes, where, by how much, and whether the trend is statistically significant**. This description is from the official challenge page.

Our answer: Terra Trend

A student, teacher, journalist or community member opens Terra Trend, picks a city, and studies a long period of environmental data without writing programming code. Instead of only showing a colorful graph, the website provides a numerical trend and explains uncertainty.

NASA question	How our website helps
What is changing?	Choose temperature, precipitation, humidity, wind speed, or solar energy.
Where?	Choose one of ten preset cities or click any place on the map.
How much?	View the estimated change per decade and over the analyzed period.
Is it significant?	Examine a two-sided slope test, p-value, and 95% confidence interval.
Can places differ?	Compare two preset locations over shared years with the same variable.

What the website does not claim

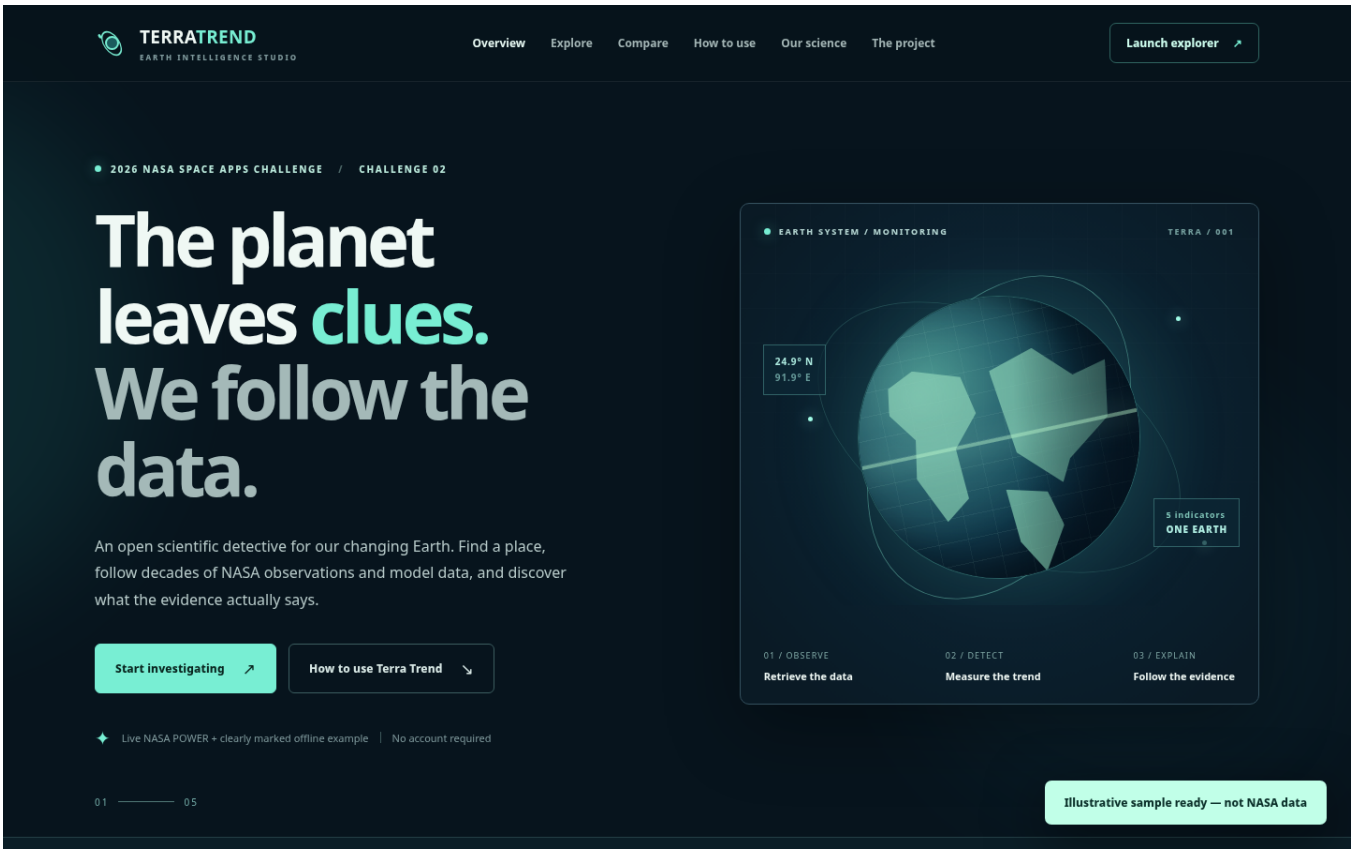
- It does not measure the atmosphere itself using a sensor in Sylhet.
- It does not predict future climate, diagnose causes, or prove human or natural causation.
- It does not automatically submit the project to NASA.

> **Beginner analogy:** Imagine a detective examining decades of dated records. The raw records are clues, the graph shows patterns, the trend gives a direction, and the significance test asks how strongly the records support that fitted direction.

02 / Visual tour and navigation

Where everything is located on the website.

The screenshot below comes from the verified website. On a smaller screen, panels may stack vertically rather than appear side by side.



Actual Terra Trend desktop screenshot (verified version).

Website tab / control	Purpose
Overview	Home: project introduction and the buttons to begin.
Explore	Main investigation: location, indicator, years, map, graph, statistics, reports.
Compare	Two locations, same measurement and shared historical years.
How to use	Eight plain-language steps and an interpretation example.
Our science	Sources, calculations, statistical method and limitations.
The project	Challenge description and copyable summary for submission.
Launch explorer	Shortcut to Explore; it does not launch a separate app.

> **Helpful navigation tip:** This is one long scrolling website with sections. Clicking a top navigation name jumps to that section. It is not opening a different website page.

03 / Your first investigation: click by click

Follow this exactly once, even if you have never studied climate science.

01. **Open the file.** Download *Terra_Trend_FINAL_Open_In_Chrome.html* and double-click it from your Downloads folder. Use Chrome or Edge. Do not rely on a file preview inside ChatGPT.
02. **Go to Explore.** Click *Start investigating* on the homepage or *Explore* in the top menu.
03. **Choose the practice mode.** Set *Data source / working mode* to *Offline interactive example - NOT NASA*. It is the safest first trial because it needs no NASA connection.
04. **Pick a location.** Choose *Sylhet, Bangladesh*. Its approximate coordinates will appear automatically.
05. **Pick a measurement.** Choose *Air temperature (°C)*. This asks about changes in average near-surface air temperature.
06. **Set years.** Use *From: 1990* and *To: 2025*. Those are 36 calendar years when every year has usable data.
07. **Click the action.** Click *Analyze sample data*. The sample plot may also update automatically after a setting change.
08. **Scroll to the evidence.** Find the chart. The colored line shows yearly values and the dashed line shows the overall fitted trend.
09. **Read the four cards.** Look at *Trend / decade*, *Estimated total change*, *Statistical test*, and *Years analyzed*.
10. **Read the report and try an export.** Scroll to *The Detective's Report*; then try *Copy findings* and *CSV*. The practice file is labeled synthetic.

> **What you should see:** A line graph, four numeric cards, and a written interpretation. In practice mode, all numbers are examples. They are **not** the real temperature history of Sylhet.

04 / Data modes and data status

This is the most important distinction in the entire project.

Mode A - Offline interactive example

This setting creates repeatable **synthetic** monthly values inside your browser. These let you check whether choosing cities, indicators, years, charts, comparisons, exports and reports works, even without internet. The page identifies these values as **SIMULATED / NOT NASA** or similar.

Mode B - Live NASA POWER

This setting requests monthly point data from the official NASA POWER service. Its success depends on the browser, network, NASA service, and, when hosted on Netlify, the website's same-origin proxy. The standalone HTML file may fail to access NASA directly because of cross-origin browser restrictions.

Label / message	How to understand it
DEMO / NOT NASA	Practice mode is selected. Treat all values as fictional examples.
NASA LIVE / ONLINE	Live mode selected; no guarantee that data have loaded yet.
Ready to connect	The site is prepared for a live request.
Requesting NASA POWER data	A live request is in progress.
NASA · LIVE	Real API response was processed by the app. Verify provenance before submission.
NASA · CACHED	A recently saved API response was reused; not a fresh network retrieval.
REQUEST FAILED	The request did not produce a usable result. Do not interpret previous/blank numbers as evidence.
SIMULATED · NOT NASA	Practice results were generated locally.

What happens when you switch modes?

The app clears older results so they cannot be mistaken for the new source. It enables or disables actions depending on whether a current analysis has finished. On a real-data error, it shows an explanation and a button to try the offline example; it does **not** silently make up NASA values.

> **Golden rule:** Every screenshot or CSV you submit as a NASA-based finding must come from verified **Live NASA POWER** mode, not from Offline example mode.

05 / Selecting places and reading the map

How the site knows which part of Earth to study.

Preset city list

The Explore dropdown currently includes ten places: Sylhet, Dhaka, New Delhi, London, New York, Cairo, Nairobi, Sydney, São Paulo and Reykjavík. Choosing one automatically fills in approximate city-center coordinates. The comparison dropdown uses these same preset locations.

Latitude and longitude - simple meaning

Coordinate	What it tells you	Valid range
Latitude	How far north (+) or south (-) of the Equator.	-90 to +90
Longitude	How far east (+) or west (-) of the reference meridian.	-180 to +180

Example: The site uses roughly 24.89° N, 91.87° E for Sylhet. These are map coordinates, not a street address or the location of a specific weather station.

Three ways to set a place

01. Select a city from the dropdown - easiest option.
02. Click anywhere on the interactive world map - the pin moves and *Custom coordinates* becomes selected.
03. Type a valid latitude and longitude yourself - also switches to *Custom coordinates*.

How the map responds

A marker highlights the selected point. Changing the point updates the location description and clears an outdated live result. The map uses a simplified world outline; selecting a point does **not** mean the website has detected the exact name of a city there.

> **Common problem:** If you type latitude 125 or longitude 220, the app cannot analyze it. Correct the number to be within the valid Earth coordinate range.

06 / The five environmental indicators

What each dropdown option means and when to choose it.

Website label	What it measures	Unit and example question
Air temperature	Typical air temperature close to the ground (about 2 metres). NASA parameter **T2M**.	**°C** - Has the location become warmer over decades?
Annual precipitation	Estimated water falling as rain and other precipitation, summed across a year. **PRECTOTCORR**.	**mm/year** - Does the yearly total tend to rise or fall?
Relative humidity	How close the air is to holding the maximum water vapor possible at that temperature. **RH2M**.	**%** - Has the typical level of humidity changed?
Wind speed	Average speed of air at about 10 metres above the surface. **WS10M**.	**m/s** - Is the wind generally stronger or weaker?
Solar energy	Average daily solar energy reaching each square metre of surface. **ALLSKY_SFC_SW_DWN**.	**kWh/m²/day** - Has available sunlight energy changed?

Which one should you select first?

Temperature is easiest for a first-time demonstration, because degrees Celsius are familiar. Next, try precipitation: yearly rainfall totals are also understandable, but note that rainfall is more variable and that annual rainfall is not a simple average of twelve monthly totals.

Understanding the units

- **°C:** a difference of +1°C is a one-degree difference, not a percentage.
- **mm/year:** an annual total of 1,000 mm means a depth of 1,000 millimetres if evenly spread over a level surface.
- **% humidity:** 75% relative humidity is not the same as “75% water in the air.”
- **m/s:** metres travelled per second by moving air.
- **kWh/m²/day:** kilowatt-hours of daily sunshine energy per square metre; this is an energy measure.

> **Important:** Do not compare the raw numerical size of a temperature trend with a rainfall trend. They use different units and describe different physical processes.

07 / Choosing years and starting the analysis

Study periods control what data is examined, and may change your conclusion.

From / To fields

The website accepts whole-year ranges beginning in **1984** and ending no later than the **last fully completed calendar year** (currently 2025 in November 2026). A request needs at least three calendar years. The app calculates only years for which all twelve monthly values are present and valid.

Choice	Meaning
1990 - 2025	Request 36 calendar years, inclusive. If some years are incomplete, fewer than 36 are analyzed.
2000 - 2010	Request 11 calendar years. This is a shorter record with potentially more variable trend results.
From 2025 to 1990	Invalid: first year must not be later than the last year.
2024 - 2025	Invalid: fewer than three requested full years.

What clicking Analyze actually does

01. Read location and check coordinate values.
02. Read indicator and validate the year range.
03. Fetch monthly NASA POWER data or create explicitly labeled practice values.
04. Omit years missing any month; convert the others to yearly values.
05. Fit a straight-line trend and calculate the test statistics.
06. Draw the graph, update four statistic cards, and write the detective report.

Why results may change with different years

A decade can be unusually hot, wet or dry. Changing start and end years may change the slope, uncertainty and significance; this does not automatically mean the tool is broken. Scientific interpretations must always say which years were studied.

> **Try this:** In the practice mode, compare the same city over 1990-2025 and 2005-2025. See whether the trend per decade changes. This is a lesson about sensitivity to the selected period, not a real-world conclusion.

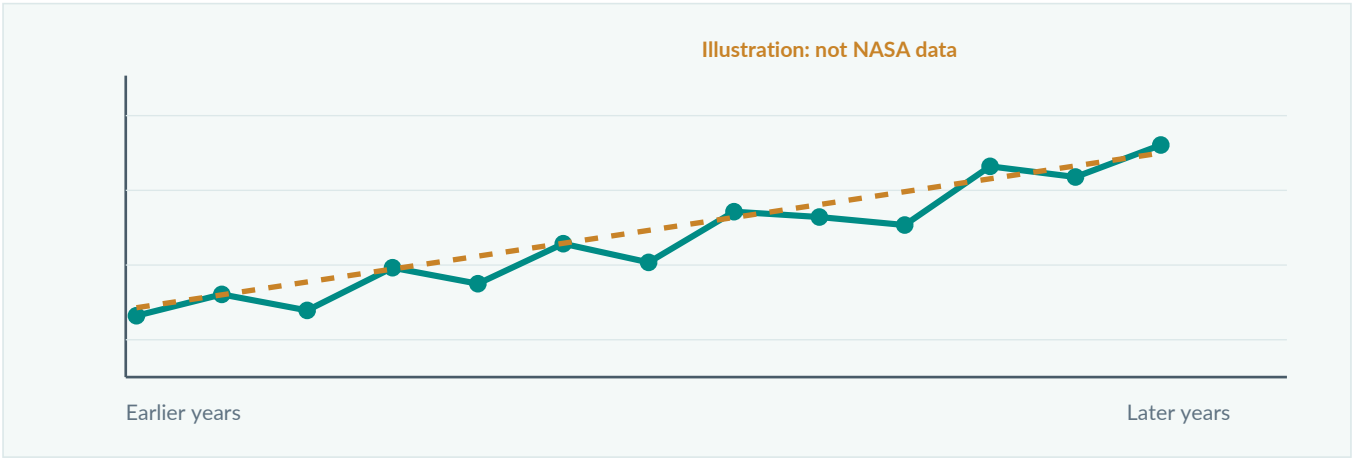
08 / Reading the trend chart

The plot is a visual summary, but it is not the whole conclusion.

Parts of the chart

What you see	Meaning
Bottom / x-axis	Calendar year (time runs left to right).
Vertical / y-axis	Value of the selected variable, in the unit displayed beside the chart.
Solid line	Connects the annual data values year by year; it may zigzag.
Dots / hover tooltip	Reveal one year and its numerical value when hovered or focused.
Dashed line	A fitted straight-line trend estimating the long-run direction.
Chart title and years	The variable, place, and actual complete years included.
Source badge	Shows whether the values are practice, fresh NASA response, or cached NASA data.

A tiny practice graph



Educational sketch: the solid line moves up and down; the dashed trend line summarizes a longer-term rise. It is not an actual Terra Trend finding.

What if the lines disagree?

A particular year may be lower than the year before, even if the fitted trend slopes upward. A trend describes the overall estimated direction across many annual points; it does not imply every single year moved in that direction.

09 / The four key result cards

These four values are the heart of your investigation.

Card	Meaning	How to explain it aloud
Trend / decade	The fitted amount of change over ten years. A plus sign means increasing and a minus sign means decreasing.	“The estimated long-term change was about X units every ten years.”
Estimated total change	The fitted yearly slope multiplied by the time between the first and last analyzed year.	“Over the measured interval, the fitted line changes by approximately X.”
Statistical test	“Detected” means p is below 0.05 for the linear slope; “Inconclusive” means the threshold is not reached.	“The evidence for a nonzero linear trend was / was not strong enough under this model.”
Years analyzed	The count of usable, fully complete annual records, which may be smaller than requested.	“The analysis used X complete calendar years.”

One fully explained hypothetical example

Suppose you selected **1990-2025** and received an estimated temperature slope of **+0.020°C per year**. Then:

Calculation	Result
Trend per decade	0.020 × 10 = **+0.20°C per decade**.
Elapsed span	2025 - 1990 = **35 years**, although the inclusive set contains 36 calendar years.
Fitted total change	0.020 × 35 = **+0.70°C**.
Years analyzed	**36** if every annual record is complete.

Do not confuse: “Estimated total change” is based on the fitted straight line; it is not necessarily the observed last-year value minus the observed first-year value. The above numbers are made up only to teach you the calculation.

10 / P-values, confidence intervals and significance

The statistics can sound intimidating, but the core idea is simple.

What is the p-value trying to answer?

The code tests a **null hypothesis**: the true linear slope is zero within the model. It uses a conventional two-sided t-test. A small p-value means that, if a zero slope and the model assumptions were true, a result as extreme as the observed one would be unusual.

Result	Responsible interpretation
p = 0.01	Below 0.05: evidence for a nonzero linear trend under the statistical assumptions.
p = 0.37	Above 0.05: this analysis does not establish a nonzero slope at the 5% threshold.
“Detected”	Means the chosen threshold was crossed - not that a problem or danger was detected.
“Inconclusive”	Does not prove the variable remained unchanged.

What does a 95% confidence interval mean?

It gives a range of plausible slope values using a standard statistical procedure. The app reports the lower and upper bounds for change **per decade** in its written detective report. If the model is appropriate, repeated intervals made this way would cover the underlying slope about 95% of the time. It is not “a 95% chance this one slope is inside this interval.”

What does statistical significance NOT mean?

- Not proof that climate change or any particular human activity caused the trend.
- Not a guarantee the trend will continue into the future.
- Not a measure of whether the change is practically important: a small change can still be statistically significant.
- Not a complete time-series model: year-to-year residual correlation can make the classical p-value too optimistic.

> **Say this during a presentation:** “We show both the estimated magnitude and the statistical uncertainty. Our results are descriptive and are not causal predictions.”

11 / The Detective's Report and export buttons

How to turn a graph into understandable text and downloadable evidence.

The Detective's Report

This section is generated after a successful investigation. It states the location, indicator, analyzed years, long-term trend, fitted total change, test result, confidence interval, and important limitations. Its title changes depending on the measurement and whether the fitted slope reached the significance threshold.

Measurement Notes

The notes identify the source mode, how the monthly data were combined into annual values, which years were incomplete, and the limits of ordinary least squares. Read them before repeating any finding in a challenge presentation.

Button / link	What it does	Practical use
CSV	Downloads a CSV table of yearly data and fitted yearly values.	Use it in Excel, Sheets, or as reproducible evidence.
Copy findings	Copies the written interpretation and source label to the clipboard.	Paste into a report or speaking notes.
Print / save PDF	Opens the browser printing interface. Select **Save as PDF** if you want a PDF of that report/page.	Create a printable record of one specific run.
Read source methodology	Opens official NASA POWER documentation in a new tab.	Understand the upstream data source.

What the CSV contains

The exported file starts with comment lines describing the source, coordinates, variable, p-value, trend and incomplete years, followed by columns such as **year**, **annual_value**, **ols_fitted_value** and **unit**. Practice CSV filenames clearly begin with *ILLUSTRATIVE-NOT-NASA*.

> **Reminder:** A CSV file saved from the website is not the original unmodified NASA file: it contains annual values computed by Terra Trend from monthly data and the fitted trend. Keep the source request URL when reporting results.

12 / Comparing two locations

The “Compare” section answers: do two places tell different stories?

Step-by-step

01. First set your indicator, years and data mode in the Explore controls.
02. Open the **Compare** tab/section.
03. Choose **Location A** - for example Sylhet.
04. Choose a **different Location B** - for example London.
05. Click **Compare evidence**.
06. Read the two lines, the per-decade trend summaries, and the written comparison.

What the chart is showing

Both places are evaluated with the **same selected variable** and the **same overlapping complete years**. If one location has missing years, the app uses the years available for both. The displayed trends may rise, fall, or differ in magnitude.

What the comparison does NOT do

The website reports a **descriptive difference in estimated slopes**, but it does **not run a statistical significance test on the difference between the two slopes**. Therefore, do not say the locations are “significantly different from each other” solely because one has $p<0.05$ and the other does not.

> **Example research question:** “Do annual temperature trends have the same direction in Sylhet and London over 1990-2025?” Run this in Live NASA mode and verify both series before stating any actual answer.

13 / The other sections of the website

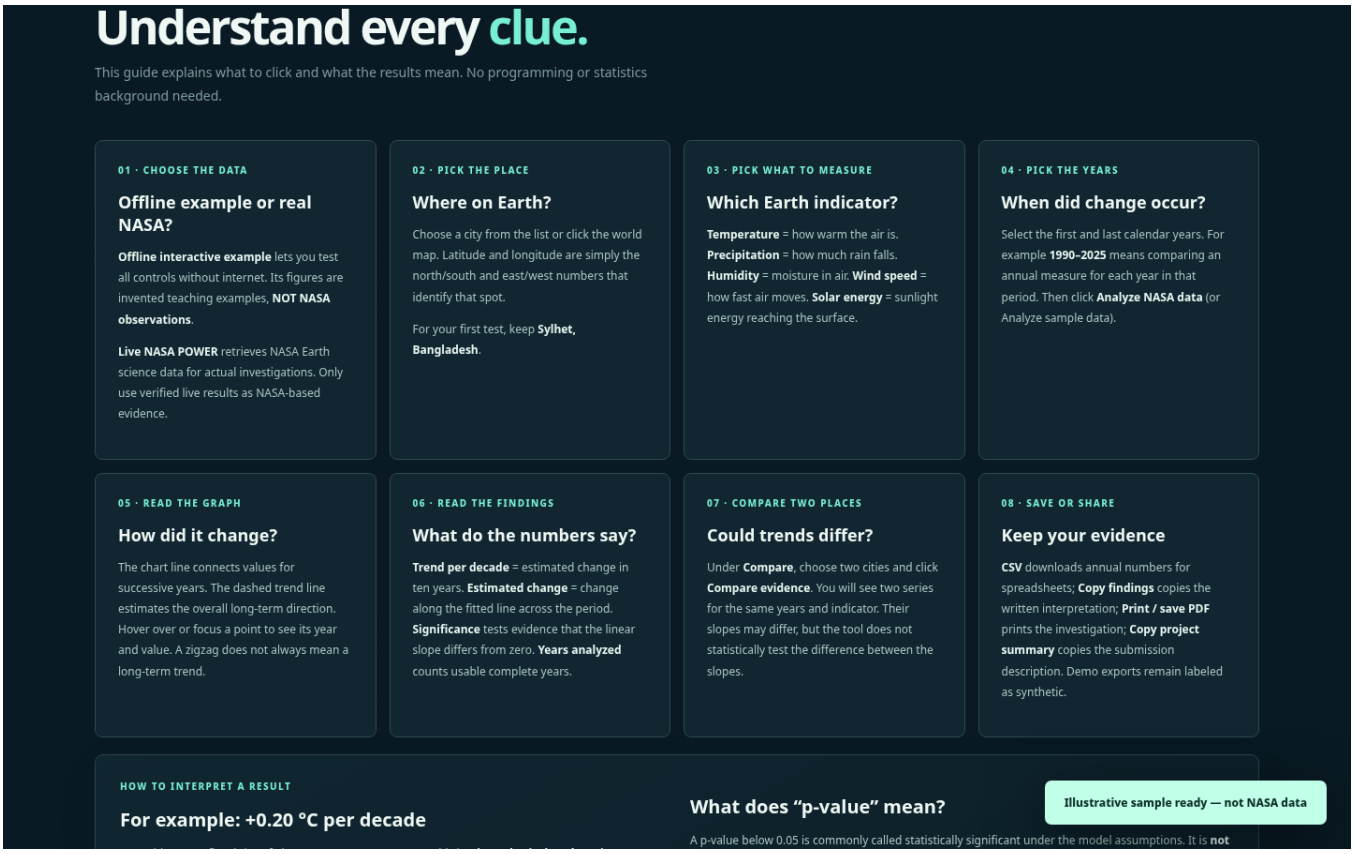
All navigation items outside Explore and Compare, explained.

Overview

The landing page introduces the Terra Trend mission. **Start investigating** and **Launch explorer** jump to the investigation form. **How to use Terra Trend** jumps to the embedded beginner tutorial. The homepage graphics are visual context; they do not claim live analysis unless a dataset has actually been loaded.

How to use

Contains eight guided steps: select source mode; choose place; choose indicator; select years; read the graph; interpret numbers; compare places; save/share results. It also includes a hypothetical example about change per decade and a short p-value explanation.



Actual screenshot of the built-in How to use section.

Our science

Explains the four-stage scientific workflow - Source, Prepare, Test, Interpret - and includes the official NASA POWER documentation link and official challenge link. It also explains the line equation and cautions about uncertainty and causation.

The project

Shows the problem, project card, chosen NASA challenge, data source, statistical method, and a **Copy project summary** button. **Official Space Apps portal** opens the actual event site. This website has no automatic project-submission feature.

14 / How NASA data reach the website

A behind-the-scenes explanation without programming jargon.

A six-stage chain of evidence

1. **Ask** Choose point, years, variable
2. **Retrieve** Request NASA POWER monthly JSON
3. **Inspect** Check valid months and skip incomplete years
4. **Summarize** Convert monthly estimates to annual values
5. **Analyze** Fit OLS line; compute trend and evidence
6. **Explain** Chart, numbers, report, comparison, CSV

What exactly is NASA POWER?

NASA POWER is a public scientific data service containing solar and meteorological estimates for geographic locations. Terra Trend uses its **Monthly Point API**, which returns month-by-month time-series data in a JSON response. This meets the project's focus on Earth science **model-derived and observationally informed** variables.

Not an exact on-site sensor

Selecting Sylhet coordinates does not mean NASA placed a thermometer at that exact point. NASA POWER uses spatially gridded datasets from products and scientific models, and conditions at a city can differ from values estimated for the relevant grid cell.

Why a hosted site helps

Modern browsers sometimes block a locally opened HTML page from reading an external API. The website publishing package contains a Netlify rewrite at **/api/power** intended to send requests to NASA through the same website origin. Whether it works on the eventual host must still be tested in reality.

15 / How the scientific calculations work

This section explains the formulas in simple words; you do not need to memorize them.

Step A - Convert 12 months into one year

A year is accepted only if the selected indicator has **all twelve valid monthly values**. Missing months, including NASA missing-value codes, cause that year to be left out. Leap years account for February having 29 days.

For temperature, humidity, wind and solar energy: the app computes a day-weighted average of the monthly values. A 31-day month gets slightly more weight than a 28-day month.

For precipitation: the app multiplies the estimated average precipitation rate in millimetres per day by the days in each month, then adds the months to obtain millimetres per year.

Step B - Fit a simple straight line

> **Fitted value = starting level + (slope × year).** The exact computer calculation is ordinary least squares (OLS), which chooses the line that minimizes the total squared differences from annual values.

The slope is in indicator-units per year. Multiply it by 10 to display a per-decade change. Multiply it by the number of years between the first and last included observations to estimate total fitted change.

Step C - Measure uncertainty

The app calculates an OLS standard error, a conventional two-sided t-test for slope = 0, an associated p-value, and a 95% confidence interval. These classical procedures assume independent errors. Environmental time series often have year-to-year dependence, so the simple calculation can overstate the precision.

Step D - Report carefully

Terra Trend reports the sign and magnitude of the slope, the usable years, a confidence interval, a significance label and a plain-language interpretation. It does not estimate causal effects or make future forecasts.

16 / A worked example you can explain aloud

All values on this page are hypothetical and are not NASA measurements.

ILLUSTRATIVE EXERCISE / NOT A FINDING

Question	Made-up example result
Where?	Sylhet (illustrative only).
What?	Annual mean air temperature.
When?	1990-2025; 36 complete years assumed.
Slope	+0.020°C/year.
Trend per decade	+0.20°C/decade.
Fitted total change	+0.70°C over 35 elapsed years.
p-value	0.01 - below the 0.05 threshold.
Confidence interval	Suppose +0.08 to +0.32°C/decade.

How to describe that result correctly

“In this **fictional teaching example**, the fitted annual temperature trend rises by about 0.20°C each decade. Across the years 1990-2025, the fitted increase is about 0.70°C. A hypothetical p-value of 0.01 would pass the 5% threshold for a nonzero slope under the regression assumptions. The example does not tell us the cause and is not real NASA evidence.”

How NOT to describe it

Do not say “NASA proved Sylhet warmed by 0.70°C,” “NASA proved climate change caused it,” or “this will rise another 0.20°C next decade” without an appropriate verified analysis and additional evidence.

> **Scientific habit:** Whenever you present a result, say **location + variable + years + estimated change + uncertainty + data source**. This makes your explanation understandable and reproducible.

17 / Ideas for a meaningful demonstration

You only need one well-explained investigation to make the project understandable.

Three questions you can investigate

Question	Suggested settings	Why it is interesting
Has annual air temperature changed in Sylhet?	Live NASA; Sylhet; T2M; 1990-2025.	Local, familiar, easy to explain.
Are precipitation trends similar in Sylhet and London?	Live NASA; PRECTOTCORR; Compare; 1990-2025.	Illustrates geographic differences and rainfall variability.
Does solar energy exhibit a trend at a location?	Live NASA; e.g. Cairo; ALLSKY_SFC_SW_DWN; 2001-2025.	Shows a second type of Earth-system variable.

How to pick a persuasive question

- Prefer one variable and a long period with complete data.
- Use a place the audience recognizes, then add a contrasting location.
- Read the data-source label and the year count before interpreting the result.
- Avoid claiming every finding is large or significant; honest inconclusive results can be useful.
- Explain why your chosen variable matters to people, ecosystems, or energy.

> **For your first presentation:** Start with Sylhet temperature; then compare Sylhet with London. The actual directions and magnitudes must be taken from a verified live run, not predicted in advance.

18 / Opening, publishing and checking the website

One local HTML file for learning; one ZIP for a public website.

Option A - Open directly, without installation

01. Download the final single-file HTML from your ChatGPT conversation.
02. Double-click the HTML in Windows File Explorer to open Chrome or Edge.
03. Select **Offline interactive example**.
04. Try the map, dropdowns, graph, comparison and CSV; no Python or terminal is needed.

Option B - Make a public website link

01. Download **Terra_Trend_VERIFIED_Ready_To_Publish.zip**.
02. Open **https://app.netlify.com/drop** in a browser. Netlify Drop currently accepts a website folder or ZIP.
03. Upload the prepared ZIP or extract it and drop the folder containing **index.html**.
04. Open the HTTPS site link ending with **netlify.app**.
05. In Explore, switch to **Live NASA POWER**, analyze Sylhet temperature for a reasonable historical period, and check the resulting source and chart.
06. Try a second location and indicator, CSV download and a comparison before using that link in your challenge submission.

Why the files differ

File	Purpose
FINAL_Open_In_Chrome.html	One self-contained file for using and reviewing the offline experience.
VERIFIED_Ready_To_Publish.zip	Website files and hosting redirect rule for a public site.
VERIFIED_Complete_Project.zip	Source code, methodology and project-submission notes as well as the website.

Critical check: A public link proves the site is hosted; it does **not** prove NASA requests work. Confirm at least one authentic Live NASA response on that published URL.

19 / Submission: what to do during Space Apps

Publishing a website and submitting an entry are separate actions.

The official submission window

As listed on the official 2026 NASA Space Apps site, the hackathon is **14-15 November 2026**; project submission opens **14 November at 9:00 AM** and closes **15 November at 11:59 PM, local event time**. Confirm your event's instructions and any rule changes before the deadline.

Your complete submission checklist

- Confirm that you are registered in the correct 2026 local event and solo Terra Trend team.
- Confirm the finalized challenge is **Be An Earth System Trend Detective!**.
- Check the challenge and local-event rules about work prepared before the event; make sure your submission is eligible.
- Have a working **public HTTPS URL** that judges can open.
- Verify real NASA POWER results, including the source request, indicator, location and years.
- Write the problem, solution, methods, data sources, limitations and significance clearly.
- If requested, provide the project code/repository and a short demo or video.
- Use your Space Apps **team page - Project tab** to make the actual submission before the closing time.
- Confirm the team page shows that the entry was submitted.

What judges should understand

What problem the website solves; why location-to-location variation matters; how NASA data enter the tool; what trend and statistical significance mean; what is innovative about the accessibility; and what scientific limitations you acknowledge.

> **Important:** Terra Trend's *The project* section can copy a project summary, but that button does **not** send anything to NASA.

20 / How to present Terra Trend confidently

You can demonstrate this project even if you are not an Earth science specialist.

A clear two-minute presentation script

Opening: “Our challenge asks us to discover how Earth-system measurements change across time and place, and whether the observed trends are statistically meaningful.”

Problem: “NASA provides valuable environmental datasets, but they can be difficult for new learners to investigate and interpret.”

Solution: “Terra Trend is an interactive web tool. A user chooses a city, an indicator, and a historical period. It retrieves NASA POWER monthly estimates and converts them into yearly evidence.”

Demo: “Here is Sylhet, the temperature indicator, and 1990-2025. We can see the annual line, the fitted trend, change per decade, total estimated change, the p-value, and the confidence interval.”

Comparison: “We can also compare a second location over the same indicator and shared years to see whether the directions and magnitudes differ.”

Responsible science: “These are gridded historical estimates; our tests describe trends rather than prove causes. We explicitly distinguish authentic NASA data from a separate educational offline mode.”

Close: “Our aim is to make scientific trend analysis understandable, transparent and reproducible for people without a technical background.”

Seven questions you might be asked

Question	Short answer
Why NASA POWER?	It provides documented historical environmental datasets via a public API.
What is a trend?	An overall estimated change across years, summarized by a fitted slope.
What is OLS?	A method that fits the straight line closest to the annual data in a squared-error sense.
What is p<0.05?	A conventional indication that the estimated slope differs from zero under the model assumptions.
Why compare locations?	The same environmental variable can behave differently in different regions.
Is this real-time weather?	No. It studies historical monthly-derived annual indicators.
Can it predict future warming?	No. It is not a forecasting or causal model.

21 / Verification status and honest limitations

What was tested, what remains uncertain, and why that matters.

What has been checked

Area	Result in the saved verification report
Main browser interactions	19 / 19 checks passed.
Simulated NASA-format responses	8 / 8 checks passed; these were intercepted, not live NASA calls.
Responsive layout and regressions	17 / 17 checks passed across mobile/desktop sizes.
Standalone science functions	Independent checks covered linear slope, leap-year precipitation and missing-month exclusion.

What has not been established

- A real, current NASA POWER request succeeding through your final public hosting URL.
- The behavior of every browser, network and future NASA API state.
- Scientific validity beyond the selected basic OLS model and its assumptions.
- Event eligibility or acceptance of prebuilt work; check event rules and organizers.

Scientific limits to say clearly

NASA POWER represents gridded environmental estimates. The website uses only selected variables and a straight-line model. It excludes incomplete years. It does not account for all physical processes, temporal autocorrelation, measurement uncertainty, or differences among data products. P-values and confidence intervals are conditional on assumptions.

> **What a failed connection means:** If the website displays a network or data error, it has not produced evidence for that query. Use the offline mode for learning, then retry Live NASA on the hosted website. Never treat the offline results as substitutes.

22 / Troubleshooting and frequently asked questions

Simple actions for common problems.

Problem	What to do
The downloaded HTML seems unresponsive.	Download it fully and open it in Chrome or Edge outside ChatGPT's preview.
The NASA chart stays empty.	Check internet and Live NASA status. On a local HTML file, publish to Netlify for the intended proxy.
“REQUEST FAILED” appears.	Read the message. A timeout, NASA error, browser restriction or missing data may have occurred. Retry with fewer years.
My city choice changes but results disappear.	This is intentional: outdated results are cleared. In Live mode, click Analyze NASA data again.
Changing a practice setting updates automatically.	Expected. Offline mode can regenerate teaching data on changes.
Custom map location shows no city name.	The app tracks coordinates; it does not reverse-geocode an exact city.
Comparison will not run.	Choose two different preset cities and valid overlapping years.
I see fewer years than requested.	One or more years lacked all twelve valid months, so they were excluded.
CSV button cannot be clicked.	Run an investigation successfully first.
Copy does not work.	Browser clipboard permissions may restrict copying, especially from local files.
Printing opens a dialog.	Choose a printer or “Save as PDF” in the browser print window.
Does the website need login?	No. It has no user accounts or saved personal profiles.

If you need to report a bug

Record the browser, source mode, chosen location, variable, years, button clicked and exact error text. That information distinguishes source-connection problems from a display problem.

23 / Glossary: scientific and website terms

A quick dictionary to revisit whenever a word is unfamiliar.

API	A way for software to request data from another service.
NASA POWER	NASA's public service for many solar and meteorological datasets.
Indicator / variable	The environmental quantity selected for analysis.
Latitude / longitude	Numbers identifying north-south and east-west position on Earth.
Time series	A sequence of values arranged by date or year.
Monthly value	A measurement estimate summarized for one calendar month.
Annual value	One summarized number for a complete calendar year.
Trend	Estimated longer-term direction of change.
Slope	Amount a fitted line changes per year.
Per decade	Per ten years.
OLS	Ordinary least squares, used to fit a straight line.
p-value	A statistic describing evidence against a zero-slope hypothesis under assumptions.
Significance	A conventional decision label based on a threshold such as 0.05.
Confidence interval	An uncertainty interval calculated using a statistical method.
Null hypothesis	Starting statistical claim tested, here a zero linear slope.
Residual	Difference between an annual data point and its fitted-line value.
Autocorrelation	Relatedness of errors across nearby years.
Gridded dataset	Environmental information estimated for areas represented by grid cells.
JSON	A structured data format returned by many APIs.
CSV	Plain-text table format usable in spreadsheets.
Cache	A temporary saved response that reduces repeat requests.
CORS	Browser restrictions affecting cross-site data requests.

Proxy	A server route that forwards a data request to another source.
Synthetic data	Numbers created for demonstrations; not real observations.
Provenance	Information identifying where data came from.
Reproducible	Someone else can follow the same settings and method to check a result.
Public URL	A web address other people can visit over the internet.

24 / Final one-page checklist and official links

Use this immediately before you show the website or submit an entry.

10 checks before any real scientific claim

01. Public site opens in a current browser.
02. Source mode is **Live NASA POWER**, not Offline example.
03. Chosen location and coordinates are correct.
04. Indicator and its physical units are correct.
05. Start and end years are stated clearly.
06. The request succeeded and source link is visible/recorded.
07. Years analyzed and incomplete-year notes are reviewed.
08. Trend per decade, total fitted change, p-value and interval are interpreted appropriately.
09. Any comparison uses matching variables and overlapping years.
10. Export/report identifies the true source and does not claim a cause or forecast.

Important official resources

2026 official Earth System Trend Detective challenge
https://www.spaceappschallenge.org/2026/challenges/earth-system-trend-detective/

NASA POWER Monthly Point API documentation
https://power.larc.nasa.gov/docs/services/api/temporal/monthly/

NASA POWER home
https://power.larc.nasa.gov/

NASA Space Apps Challenge 2026 overview and dates
https://www.spaceappschallenge.org/2026/

Netlify Drop (website publishing)
https://app.netlify.com/drop

Netlify Drop official quickstart
https://docs.netlify.com/start/quickstarts/netlify-drop-quickstart/

> **End of guide.** Start with one offline practice run to understand the controls. For an actual challenge finding, publish the website, verify Live NASA POWER, record the source and years, then explain the result with its uncertainty.